MIL-T-43779C
25 February 1988
SUPERSEDING
MIL-T-43779B
31 March 1983

MILITARY SPECIFICATION

TIMER, DELAY: RELEASES, CARGO PARACHUTE, M1 and M2

* This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of mechanical delay timer assembly to be used with either the M1 or M2 cargo parpachute release.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, arid Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-676 - Boxes Setup
PPP-C-1797 - Cushioning Material, Resilient, Low Denisty, Unicellular, Polypropylene Foam

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
* MIL-STD-130 - Identification Marking of U.S. Military Property
* MIL-STD-147 - Palletized Unit Loads
* MIL-STD-731 - Quality of Wood Members for Containers and Pallets
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

DRAWINGS

* U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

11-1-894 Timer Delay; Release, Cargo Parachute, M1 and M2
* 11-1-2614 Timer and Stem Assembly
* 11-1-2660 Mounting Block Assembly
* 11-1-2666 Housing Assembly

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

* 2.2 Other publications. The following documents form part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of solicitation.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for (Commercial Packaging

* (Application for copies should be addressed to the American Society for Testing and Materials, 1619 Race Street, Philadelphia, PA 19103.)

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* (Technical society and technical associations documents are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Materials and construction. The materials (see 6.4), construction and supplementary processes involved in the fabrication of the delay timer assembly shall conform to the requirements specified herein and as specified on Drawings 11-1-894, 11-1-2614, 11-1-2660, 11-1-2666, and all subsidiary drawings pertaining thereto.

3.3 Performance.

3.3.1 Timing cycle. The delay timer timing cycle shall be 12 seconds, plus or minus 3 seconds, from start till completion, and there shall be no evidence of interference in dropping down to its unlocked position after completion of the unwinding cycle when tested as specified in 4.5.1.

3.3.2 Delay timer keys. When functionally tested as specified in 4.5.2, the delay timer keys shall protrude a minimum of 5/32 inch from each side of the housing during the timing cycle. At completion of the timing cycle, the keys shall protrude not more than 1/32 inch from each side of the housing.

3.4 Government loaned property (GLP). Either one of the items listed below is required for the functional test (see 4.5.1 and figure 1) and may be obtained from the procuring activity on a loan basis (see 6.2).

Item	Part No	NSN
Release, Cargo Parachute, Airdrop; M-1	11-1-1487-1	1670-01-097-8816
Release, Cargo Parachute, Airdrop; M-2	11-1-565-1	1670-01-097-8817

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3.4.1 Timer and stem assembly. When the timer and stem assembly is procured separtely by the Government, a Release, Cargo Parachute, M-1 or M-2 and Timer, Delay components will be supplied (GLP) in order to perform testing as specified in 4.5.1.

3.5 Identification of product. Identification marking of the Timer, Delay and components shall be in accordance with MIL-STD-130.

* 3.6 Workmanship. All components and assemblies of the delay timer assembly shall be free from dirt and other extraneous material, burrs, slivers, dents, and cracks. External surfaces shall be free from sharp edges arid corners. Components shall not be fractured, buckled, loose, dented, missing, malformed, or misaligned. Screws shall be properly and securely seated. The timer delay releases shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order. the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herin, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirement.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3, and tested for the characteristics specified in 4.4.4. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of timers. The sample unit shall be one timer. The inspection level shall be III and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.0 for major defects and 4.0 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Omitted or not as specified	x	
	Any evidence of corrosion	x	
	Dirt, oil, grease, or other foreign matter		x
Construction and workmanship	Any component cracked, malformed, misaligned or damaged	x	
	Sharp edges, burrs, or metal slivers		x
	Components incorrectly assembled, loose, or missing	x	
	Screws improperly or not securely seated		x
Identification marking	Missing, incomplete illegible, improper location, or not as specified		x

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* 4.4.3 End item dimensional examination. The end items shall be examined for all dimensions specified on Drawing 11-1-894. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of completed delay timer assemblies. The sample unit shall be one completed delay timer assembly. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.0

* 4.4.4 End item testing. The timer shall be tested as specified in 4.5.1 and 4.5.2 for conformance to the requirements specified in 3.3.1 and 3.3.2 respectively. The lot size shall be expressed in units of timers. The sample unit shall be one timer. For each test, the inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed-in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container,
Content	Number per container is more or less than required.

* 4.5 Methods of inspection.

* 4.5.1 Delay timer unwinding cycle test. Testing of the delay assembly shall be conducted in a temperature environment conforming to the general test performance guidance requirements and Procedure I of Methods 501.2 and 502.2 of MIL-STD-810. Stabilization temperature intervals shall not exceed 2 hours. The delay assembly shall be tested with the stem and keys in a horizontal position and held in a cargo parachute release; type M-1 or M-2 (GLP, see 3.4), in the manner shown on figure 1. An additional weight shall be suspended from the delay assembly using some suitable textile or metal sling (see figure 1).

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The stem shall be fully wound in a clockwise direction approximately 90 degrees, engaging the keys in the release cavity slots (locked position), and the arming wire inserted. The assembly shall then be actuated by removal of the arming wire and the length of the timing cycle shall be measured by a suitable time lapse instrument. After the timing cycle has been completed, the delay assembly shall drop freely and smoothly within the release to its lower unlocked position. Failure of the assembly to meet either the time requirement of 3.3.1 or any indication of interference in dropping into the unlocked position, shall constitute a defect.

4.5.2 Key movement test. The delay timer shall be rewound and measurements taken of the key protrusion from the housing while in the extended position. With the keys held hanging in a vertical position, each key shall be measured for protrusion from the housing with key in a withdrawn position. Any deviation from the requirements specified in 3.3.2 shall constitute a defect.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each delay timer shall be placed in a close-fitting, clear polyethylene film bag. The bag shall be fabricated from polyethylene film having a thickness of 0.0015-inch (20 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are scraight, continuous, and parallel to each other and the formed edges of the bag. The bag closure shall be effected by heat sealing or by means of a mechanical tie (paper or vinyl plastic covered soft steel wire, aluminum bag, etc.). Each bagged delay timer shall be completely wrapped in not less than l/4 inch thickness of cushioning material conforming to type II, class B of PPP-C-1797. Alternatively each timer shall be packaged in a heat sealables plastic-cushioned, Kraft envelope. The cushioning shall have a nominal thickness of 3/16 inch. The closure shall be effected by heat sealing. Each timer shall then be unit packed in a close-fitting box conforming to variety 2, style III, type G, class i of PPP-B-566; type I, variety 2, class A, style 4 of PPP-B-676; or style RSC, type CF, grade W6c of PPP-B-636. Each box shall be securely closed with pressure-sensitive tape in accordance with the appendix of the applicable specification.

5.1.1.1 Timer and stem assembly when procured separately (see 6.2). Each timer and stem assembly shall be placed in a close-fitting, clear polyethylene film bag. The bag shall be fabricated from polyethylene film having a thickness of 0.0015-inch (20 percent tolerance). The polyethylene bag shall be formed with heat sealed seams that are straights continuous and parallel to each other and the formed edges of the bag. The bag closure shall be effected by heat sealing or by means of a mechanical tie (paper or vinyl plastic covered soft steel wires aluminum bank, etc.). Each bagged timer and stem assembly shall be completely wrapped in not less than 1/4 inch thickness of cushioning material conforming to type II, class B of PPP-C-1797. The cushioning material shall be

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secured with pressure-sensitive tape. Alternatively, each timer and stem assembly shall be packaged in a heat-sealable, plastic-cushioned, Kraft envelope. The cushioning shall have a nominal thickness of 3/16 inch. The closure shall be effected by heat sealing. Each bagged and cushioned timer and stem assembly shall then be unit packed in a close-fitting box conforming to variety 2, style III, type G, class i of PPP-B-566; type I, variety 2, class A, style 4 of PPP-B-676; or style RSC, type CF, grade W6c of PPP-B-636. Each box shall be securely closed with pressure-sensitive tape in accordance with the appendix of the applicable box specification.

* 5.1.2 Commercial. Each delay timer and separately procured timer and stem assemblies (see 6.2), shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Sixty delay timers or sixty timer arid stem assemblies, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type, style A or 1, grade A or B, type 2 load of PPP-B-601. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

5.2.2 level B packing. Sixty delay timers or sixty timer and stem assemblies, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

* 5.2.3 Commercial packing. Delay timers or separately procured timer and stem assemblies, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Harking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in. accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The delay timer assembly is used to arm the cargo parachute release (M-1 or M-2) so that it will automatically disengage the parachute from the load at the moment of ground impact to prevent cargo damage.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3 and 6.3).
c. Source and delivery of Government loaned equipment (see 3.4).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
e. When timer and stem assembly are procured separately (see 5.1.1.1 and 5.1.2).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.2.1 Government loaned property. The contracting officer should arrange to loan the property listed in 3.4.

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection and approval of the first article.

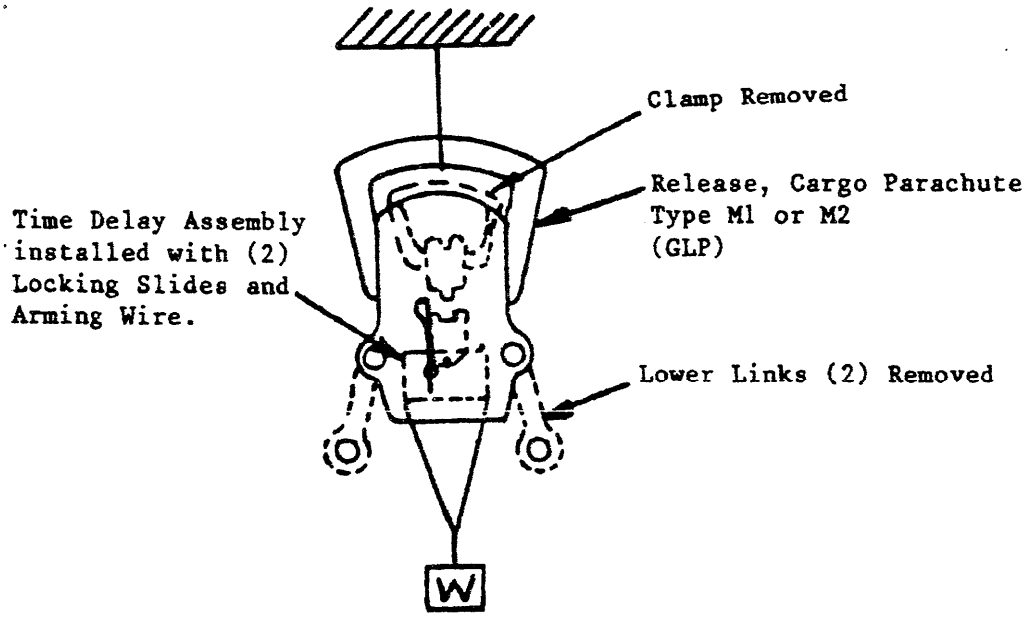
6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document {see 3.2).

* 6.5 Subiect term (key word) listing.

Release
Timer, delay

* 6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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NOTE: Total Wgt of Timer
+ 2 Slides + suspended
Wgt = 35 oz. min.

Figure - 1

ARRANGEMENT
F O R
WEIGHTED FUNCTIONAL TEST
F O R
TIME EDLAY ASSEMBLY

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-43779C	Timer, Delay: Releases, Cargo Parachute, M1 and M2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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